# Guidelines For Manufacturing Remotely Piloted Aircraft Systems In India

Although Manufacturers/OEMs of remotely piloted aircraft system(s) (RPAS) are not regulated through the Civil Aviation Requirements (CAR) Section 3, Series X, Part I, being a responsible entity, a manufacturer is expected to follow the procedure specified hereunder.

1. Apart from the indigenous manufacturers, the entity importing parts and assembling remotely piloted aircraft (RPA) in India (assembler) will also be considered as a manufacturer.

2. Manufacturers should ensure that their RPAs meet the minimum standards specified in the CAR.

3. Manufacturers should carry out necessary tests, as many as required. They may use the test sites specified in the CAR for carrying out such tests.

4. OEMs/manufacturers may alternatively utilise unused airstrips or government educational institution's campus, provided adequate safety precautions are in place. However, they need to ensure that no manned or unmanned aircraft is flying during such operations in the intended test area.

5. The manufacturers/OEMs of Nano RPA should prominently engrave/display the manufacturer's serial number. Also, they should appropriately geo-fence their nano RPA for 15 metres (50 feet) above ground level (AGL) ceiling.

6. The manufacturer/assembler of micro and above should comply with the NPNT (No Permission-No Takeoff) specifications.

7. If the RPA developed is in the nano or micro category, the minimum standards set by manufacturer/OEM will be considered. However, for the micro category, the manufacturer should certify compliance of the NPNT, equipment, and other standards specified in the CAR.

8. If the model developed is in the small or above category, the OEM/manufacturer should use the checklist for the manufacture of small and above categories of RPAS (both Indian and foreign) and get it certified by certifying agency by showing compliance to applicable standards. In addition, the manufacturer should submit a certificate of compliance.

9. Post-certification (by self or certifying agency), all models will be listed on the Digital Sky Platform with all requisite details. This will ensure that the user/operator/remote pilot picks the correct model from the list.

10. The manufacturer/OEM may obtain Equipment Type Approval (ETA) from Wireless Planning and Coordination Wing (WPC), DOT to facilitate the end-user, as the ETA is given for a particular type/model or lot in case of import.

11. The manufacturer should develop the following manuals: (a) RPA Flight Manual/ Manufacturer's Operating Manual specifying operating conditions/limitations. (b) Maintenance manual/guidelines/procedure. (c) Maintenance inspection schedule/overhaul interval. (d) Self-explanatory information booklet for end-users.

12. The manufacturer should provide the self-explanatory information booklet to end-users in the RPAS package/box.

13. Maintenance and repair of RPAS should be carried out in accordance with manufacturers' approved procedures in authorised service centres of the manufacturer/OEM or remote pilot/operator. In the latter case, the OEM/manufacturer should ensure that remote pilot/operator does not have unauthorised access to mandatory equipment/firmware (including NPNT).

14. Nano RPA Manufacturers/OEMs must ensure the traceability of the purchaser for every drone. Their agents/distributors must collect and safely store IDs at point of sale. Manufacturers should ensure that a record of serial numbers by distributor/agent is maintained and kept up to date. EFY

*Source: https://dgca.gov.in/digigov-portal*

*Latest guidelines are now available at https://www.civilaviation.gov.in/sites/default/files/Draft_UAS_Rules_2020.pdf*